DSA 8420 Spring 2025
Homework 9

Due April 1, 2025

1. (10 points) Solve the following problems by the dual simplex method.

$$\begin{aligned} \max \quad & -8x_1 - 2x_2 - 12x_3 \\ s.t. \quad & -2x_1 - x_2 - 4x_3 \leq 1 \\ & -x_1 + 2x_2 - 2x_3 \leq -6 \\ & - x_2 + x_3 \leq 2 \\ & x_1,\ x_2,\ x_3 \geq 0 \end{aligned}$$

2. (10 points) Use the dual simplex method to show that the following problem is infeasible.

$$\begin{aligned} \max \quad & -8x_1 - x_2 - x_3 \\ s.t. \quad & -2x_1 - x_2 \leq -20 \\ & 2x_1 - x_2 + x_3 \leq -12 \\ & x_1 + x_3 \leq 7 \\ & -x_1 + x_2 - x_3 \leq -5 \\ & x_1,\ x_2,\ x_3 \geq 0 \end{aligned}$$

3. Consider the following resource allocation problem and the accompanying optimal tableau, where x_4, x_5 and x_6 are the slack variables for

constraints 1 through 3, respectively.

$$
\begin{array}{llr}
\max\ z = 10x_1 + 7x_2 + 6x_3 & & \text{(Profit \$)} \\
s.t. & 3x_1 + 2x_2 + x_3 \le 36 & \text{(Resource 1)} \\
& x_1 + x_2 + 2x_3 \le 32 & \text{(Resource 2)} \\
& 2x_1 + x_2 + x_3 \le 22 & \text{(Resource 3)} \\
& x_1,\ x_2,\ x_3 \ge 0 &
\end{array}
$$

	z	x_1	x_2	x_3	x_4	x_5	x_6	RHS
z	1	3	0	0	1	0	5	146
x_2	0	1	1	0	1	0	-1	14
x_5	0	-2	0	0	1	1	-3	2
x_3	0	1	0	1	-1	0	2	8

The following three questions are independent of each other.

(a) (10 points) Suppose that the profit of producing one unit of Product 3 goes down from \$6 to \$3. Use sensitivity analysis to find a new production plan for the company.

(b) (10 points) Suppose that the available amount of Resource 1 increases from 36 units to 41 units, and the available amount of Resource 3 decreases from 22 units to 20 units. How should the production plan be changed?

(c) (10 points) Due to improvement of technology, the amount of Resource 1 needed in order to produce one unit of Product 1 is reduced from 3 units to 2 units. How should the production plan be changed?